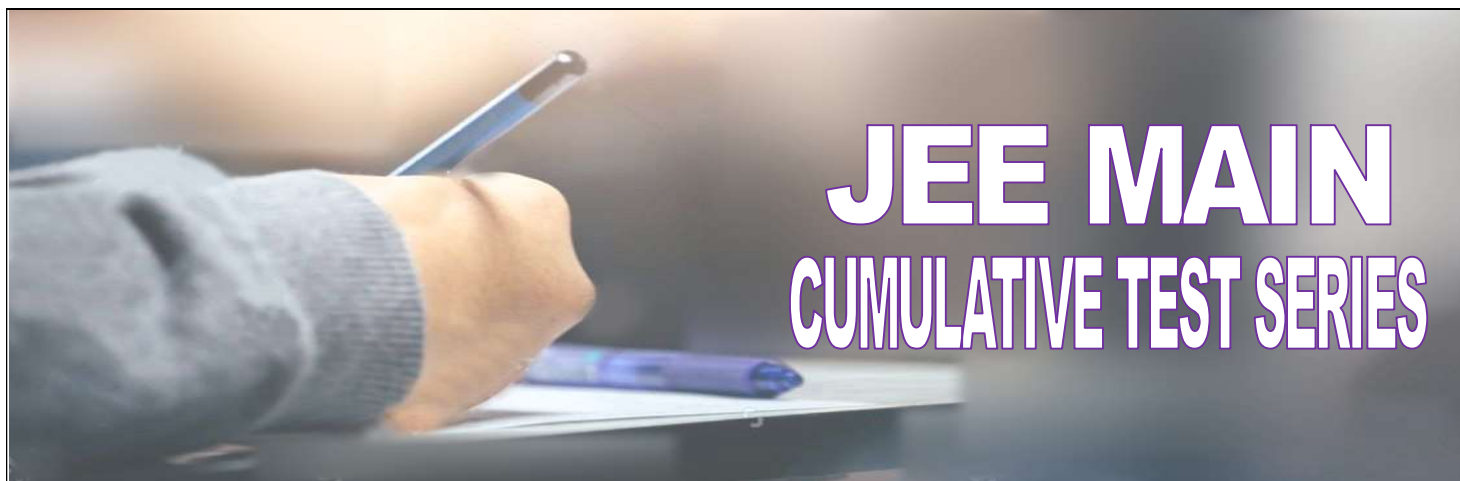




NARAYANA
IIT ACADEMY
INDIA



Sec: SR_*CO-SC(MODEL-B)

Date: 30-06-24

Time: 3 Hrs

Max. Marks: 300

30-06-24_SR.IIT_STAR CO-SC(MODEL-B)_JEE MAIN_CTM-46_SYLLABUS

PHYSICS:

PRESENT WEEK(60%): OPTICS-III : Refraction at curved surface, Lenses and combination of lenses

MAINS EXPERIMENTS: The focal length of Convex lens, using the parallax method.

ADVANCED EXPERIMENTS: focal length of a concave mirror and a convex lens using u-v method

PREVIOUS WEEK(20%): OPTICS-II : Refraction at plane surface, TIR and Prism, dispersion and deviation in thin prism

MAINS EXPERIMENTS: The plot of the angle of deviation vs angle of incidence for a triangular prism.

Refractive index of a glass slab using a travelling microscope.

DESIGNATED TOPIC(20%): Alternating Current

CHEMISTRY:

PRESENT WEEK(60%): Colour, Magnetic properties, Catalytic properties, alloy and interstitial compounds formation, oxides and oxoanions of metals, Preparation & Properties of Potassium dichromate, F-BLOCK

PREVIOUS WEEK(20%): d-Block elements : General Characteristics, electronic configuration, atomic & ionic size, density, ionisation enthalpy, melting points, enthalpy of atomization, Oxidation states, Chemical reactivity and E0 values, Nature of oxides

DESIGNATED TOPIC(20%): CHEMICAL & IONIC EQUILIBRIUM

MATHEMATICS:

PRESENT WEEK(80%): Permutations & Combinations: Fundamental Principles of counting, Linear permutations, Circular permutations

PREVIOUS WEEK(20%): Binomial theorem : Binomial theorem for a positive integral index, Constant term, Numerically greatest term, Properties of the Binomial coefficients, Multinomial Theorem

DESIGNATED TOPIC(20%): I) Triangular Inequalities(1qs)

II) Rotation conics theorem(1qs) III) Geometrical interpretation(2qs)

IV) Cube roots of unity(1qs) V) nth roots of unity(1qs)

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

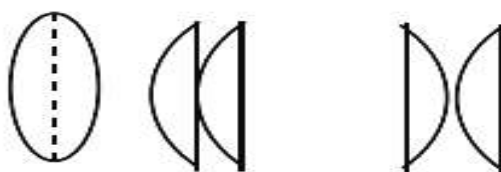
1. The two light beams having intensities I and $9I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\frac{\pi}{2}$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be:

A) $2I$ B) $6I$ C) $5I$ D) $7I$

2. Two coherent sources of light interfere. The intensity ratio of two sources is $1 : 4$. For this interference pattern if the value of $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}}$ is equal to $\frac{2\alpha + 1}{\beta + 3}$, then $\frac{\alpha}{\beta}$ will be:

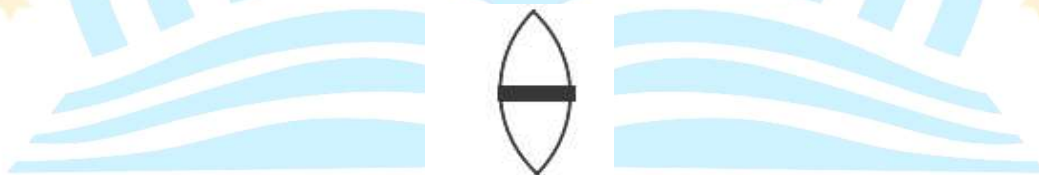
A) 1.5 B) 0.5 C) 2 D) 1

3. Two similar plano-convex lenses are combined together in three different ways as shown in the adjoining figure. The ratio of the focal lengths in three cases will be



A) $2 : 2 : 1$ B) $1 : 1 : 1$ C) $1 : 2 : 2$ D) $2 : 1 : 1$

4. If the central portion of a convex lens is wrapped in black paper as shown in the figure



- A) No image will be formed by the remaining portion of the lens
B) The full image will be formed but it will be less bright
C) The central portion of the image will be missing
D) There will be two images each produced by one of the exposed portions of the lens

5. A thin convex lens made from crown glass $\left(\mu = \frac{3}{2}\right)$ has focal length f . When it is measured in two different liquids having refractive indices $\frac{4}{3}$ and $\frac{5}{3}$, it has the focal lengths f_1 and f_2 respectively. The correct relation between the focal lengths is:

- A) f_1 and f_2 both become negative
 B) $f_1 = f_2 < f$
 C) $f_1 > f$ and f_2 becomes negative
 D) $f_2 > f$ and f_1 becomes negative

6. In an experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v , from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x-axis meets the experimental curve at P. The coordinates of P will be

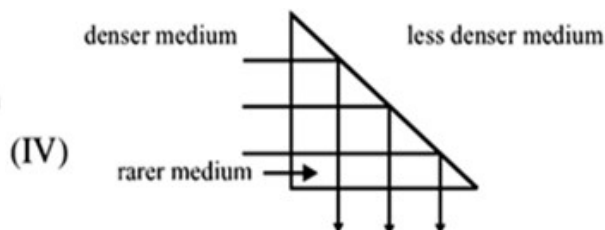
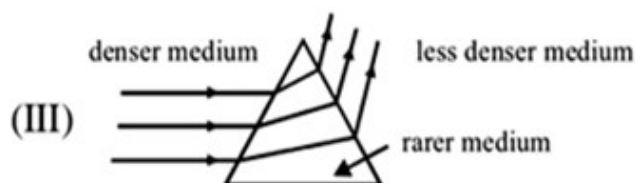
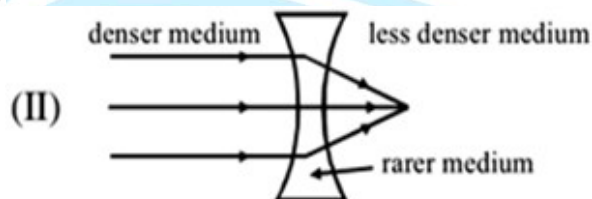
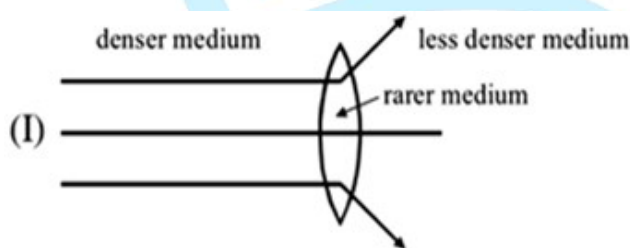
- A) $\left(\frac{f}{2}, \frac{f}{2}\right)$
 B) (f, f)
 C) $(4f, 4f)$
 D) $(2f, 2f)$

7. **Assertion (A):** If the source of light in YDSE is changed from red to violet, then the separation between the interference fringes will increase.

Reason (R): The wavelength of violet light is smaller than the wavelength of red light

- A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 C) If Assertion is True but the Reason is False.
 D) If both Assertion & Reason are False.

8. When light rays pass from an optically denser medium through an optically less dense medium as shown below, which is/ are possible?



- A) I, II only
 B) III, IV only
 C) I, II, III only
 D) IV only

9. Four combinations of two thin lenses are given in List I. The radius of curvature of all curved surfaces is r and the refractive index of all the lenses is 1.5. Match lens combinations in List I with their focal length in List II and select the correct answer using the code given below the list.

List-I



List-II

(A) $2r$

(B) $\frac{r}{2}$

(C) $-r$

(D) r

Code:

A) P – A, Q – B, R – C, S – D

B) P – B, Q – D, R – C, S – A

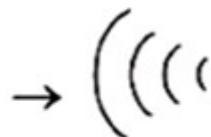
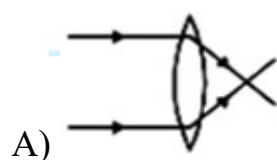
C) P – D, Q – A, R – B, S – C

D) P – B, Q – A, R – C, S – D

10. In Column I the parallel rays are incident on some optical device and in Column II the shape of emergent wave fronts are shown. Select incorrect match.

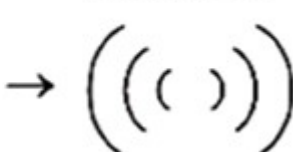
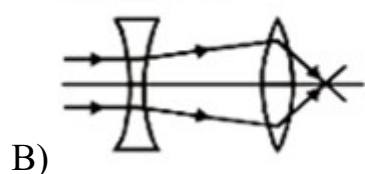
Column - I

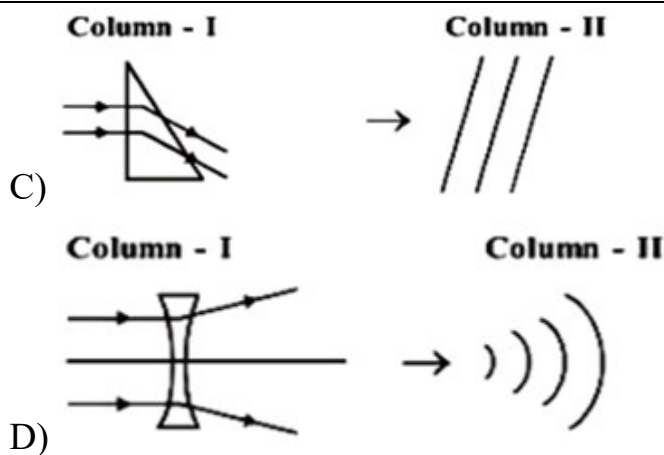
Column - II



Column - I

Column - II

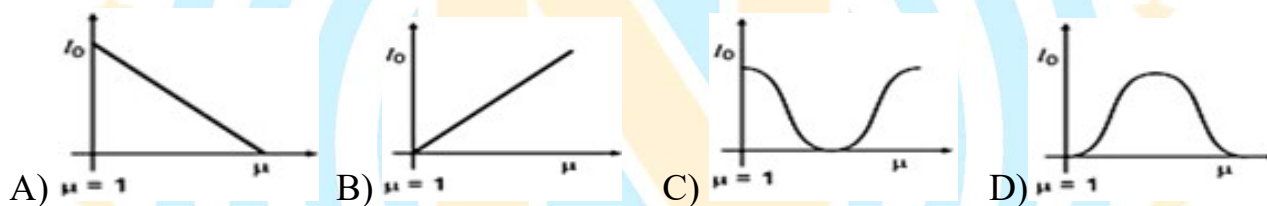




11. Let S_1 and S_2 be the two slits in Young's double slit experiment. If central maxima is observed at P and angle $S_1PS_2 = \theta$ (θ is small) find the y – coordinates of the 3rd minima assuming the origin at the central maxima. (λ = wavelength of monochromatic light used).

A) $\pm \frac{2\lambda}{\theta}$ B) $\pm \frac{5\lambda}{2\theta}$ C) $\pm \frac{5}{2} \lambda \theta$ D) $\pm 2\lambda \theta$

12. In YDSE if a slab whose refractive index can be varied is placed in front of one of the slits. Then, the variation of resultant intensity at mid – point of screen with μ will be best represented by (μ is greater than or equal to 1)



13. White light is incident normally on a glass plate (in air) of thickness 500 nm and refractive index of 1.5. The wavelength (in nm) in the visible region (400 nm. 700 nm) that is strongly reflected by the plate is
- A) 450 B) 600 C) 400 D) 500
14. Two waves from coherent sources meet at a point in a path difference of Δx . Both the waves have same intensities. Match the following two columns.

COLUMN-I	COLUMN-II
(a) If $\Delta x = \lambda/3$	(p) Resultant intensity will become three times
(b) If $\Delta x = \lambda/6$	(q) Resultant intensity will remain same
(c) If $\Delta x = \lambda/4$	(r) Resultant intensity will become two times
(d) If $\Delta x = \lambda/2$	(s) Resultant intensity will become zero

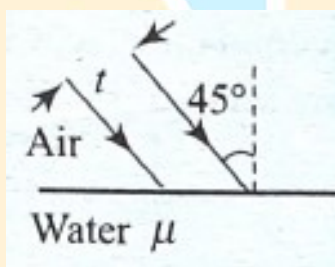
A) a – q, b – p, c – r, d – s

B) a – p, b – r, c – q, d – s

C) a – s, b – p, c – q, d – s

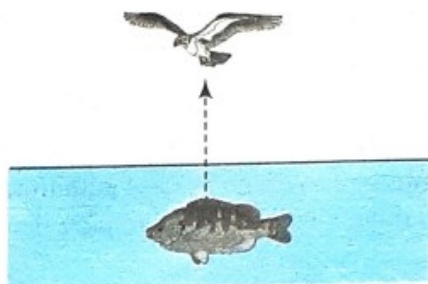
D) a – p, b – q, c – r, d – s

15. Interference was observed in interference chamber where air was present, now the chamber is evacuated, and if same light is used, a careful observer will see
 A) No interference
 B) Interference with brighter bands
 C) Interference with dark bands
 D) Interference fringe with larger width
16. Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly.
 A) 1.59
 B) 1.69
 C) 1.78
 D) 1.25
17. The lateral shift of a light ray while it passes through a parallel glass slab of thickness 10 cm placed in air, is (The angle of incidence in air is 60° and the angle of refraction in glass is 45°)
 A) 6.66 cm
 B) 3.66 cm
 C) 6.33 cm
 D) 5.66 cm
18. A beam of width ' t ' is incident at 45° on an air-water boundary. The width of the beam in water is



- A) $\frac{t \sqrt{2\mu^2 - 1}}{\mu}$
 B) $\frac{t \sqrt{\mu^2 - 1}}{\mu}$
 C) $\frac{t \sqrt{2\mu^2 - 1}}{2\mu}$
 D) $\frac{t \sqrt{\mu^2 - 1}}{2\mu}$

19. A fish rising up vertically toward the surface of water ($\mu_w = \frac{4}{3}$) with speed 3 ms^{-1} observes a bird diving down vertically towards it with speed 12 ms^{-1} . The actual velocity of bird is



- A) 3.5 ms^{-1}
 B) 5.4 ms^{-1}
 C) 6.75 ms^{-1}
 D) 3.4 ms^{-1}

20. The x-z plane separates two media A and B of refractive indices $\mu_1 = 1.5$ and $\mu_2 = 2$. A ray of light travels from A to B . Its directions in the two media are given by unit vectors $\vec{u}_1 = a\hat{i} + b\hat{j}$ and $\vec{u}_2 = c\hat{i} + d\hat{j}$. Then

A) $\frac{a}{c} = \frac{4}{3}$

B) $\frac{a}{c} = \frac{3}{4}$

C) $\frac{b}{d} = \frac{4}{3}$

D) $\frac{b}{d} = \frac{3}{4}$

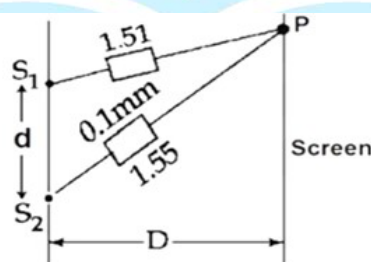
SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

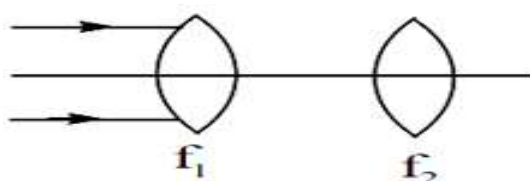
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only**. Have to Answer any 5 only out of 10 questions and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

21. In a Young's double slit experiment, an angular width of the fringe is 0.35° on a screen placed at 2 m away for particular wavelength of 450 nm. The angular width of the fringe, when whole system is immersed in a medium of refractive index $7/5$, is $\frac{1}{\alpha}$. The value of α is _____
22. In Young's double slit experiment, two slits S_1 and S_2 are 'd' distance apart and the separation from slits to screen is D (as shown in figure). Now if two transparent slabs of equal thickness 0.1 mm but refractive index 1.51 and 1.55 are introduced in the path of beam ($\lambda = 4000 \text{ \AA}$) from S_1 and S_2 respectively. The central bright fringe spot will shift by _____ number of fringes.



23. A parallel beam of light is incident on the system of two convex lenses of focal lengths $f_1 = 20\text{cm}$ and $f_2 = 10\text{cm}$. What should be the distance between the two lenses so that rays after refraction from both the lenses pass undeviated?



24. Monochromatic light of wavelength 900 nm is used in a young's double slit experiment. One of the slits is covered by a transparent sheet of thickness 1.8×10^{-5} m made of material of refractive index 1.6. How many fringes will shift due to introduction of the sheet?
25. A Young's double – slit experiment is performed using monochromatic light of wavelength λ . The intensity of light at a point on the screen, where the path difference is λ , is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{6}$ is given by $\frac{nK}{12}$, where n is an integer. The value of n is ____
26. Two slits 4.0×10^{-6} m apart are illuminated by light of wavelength 600 nm. The highest order fringe in the interference pattern is ____
27. Consider an interference experiment using eight equally spaced slits. The smallest phase difference in the waves from adjacent slits such that the resultant wave has zero amplitude is ____ (in degrees)
28. In young's double slit experiment, the slits are 2 mm apart and are illuminated by photons of two wavelengths $\lambda_1 = 12000 \text{ \AA}$ and $\lambda_2 = 10000 \text{ \AA}$. At what minimum distance from the common central bright fringe on the screen 2 m from the slit will a bright fringe from one interference pattern coincide with a bright fringe from the other?
29. A ray of light passes through a medium whose refractive index varies with distance as $n = n_0 \left(1 + \frac{x}{2a}\right)$. If ray enters the medium parallel to x-axis, then the time taken for ray to travel between $x = 0$ and $x = a$ is $\frac{(p) n_0 a}{(q) c}$. The value of $p+q$ is
30. A vessel is filled with a non-homogeneous liquid whose refractive index varies with the depth y from the free surface of liquid as $\mu = \left(1 + \frac{2y}{H}\right)$. H is the height to which the liquid is filled in the vessel. Then the apparent depth as seen by an observer from above is $\frac{H}{2} \log_e X$. The value of X is.....

CHEMISTRY

MAX.MARKS: 100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. The IUPAC name of $[CO(NH_3)_6][Cr(C_2O_4)_3]$ is:
- A) Hexaamminecobalt(III) tris(oxalato)chromate(III)
 B) Hexaamminecobalt(III) tris(oxalato)chromium(III)
 C) Hexaamminecobalt(II) tris(oxalato)chromium(III)
 D) Hexaamminecobalt(III) trioxalatechromium(III)
32. In the complex $FeK_2[Fe(CN)_6]$
- A) Both Fe atoms are in the same oxidation state
 B) Both Fe atoms are in different oxidation state
 C) The coordination number of iron is 4
 D) The complex is a high spin complex
33. The structure of $K[PtCl_3(C_2H_4)]$ and hybridization of Pt respectively are
- A) square planar, sp^3d^2 B) square planar, dsp^2
 C) tetrahedral, sp^3 D) octahedral, d^2sp^3
34. Which of the following is an outer orbital complex?
- A) $[Fe(CN)_6]^{4-}$ B) $[Mn(CN)_6]^{4-}$ C) $[Cr(NH_3)_6]^{3+}$ D) $[Ni(NH_3)_6]^{2+}$
35. On treatment of 100ml of 0.1 M solution of complex $CoCl_3 \cdot 6H_2O$ with excess of $AgNO_3$, 4.305g of $AgCl$ was obtained. The complex is (atomic weight of $Ag=108g \cdot mol^{-1}$)
- A) $[Co(H_2O)_3Cl_3] \cdot 3H_2O$ B) $[Co(H_2O)_4Cl_2]Cl \cdot 2H_2O$
 C) $[Co(H_2O)_5Cl]Cl_2 \cdot H_2O$ D) $[Co(H_2O)_6]Cl_3$
36. The pair of complex ions in which EAN of central metal is same:
- A) $[Ni(CO)_4]$, $[Fe(CN)_6]^{4-}$ B) $[Ni(en)_2]^{2+}$, $[Fe(H_2O)_6]^{2+}$
 C) $[Co(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$ D) $[Ni(en)_2]^{2+}$, $[Fe(H_2O)_6]^{3+}$

37. The oxidation state of central metal ion and its magnetic moment in brownring complex is:
 A) +1 and 2.8 BM B) +1 and 3.87 BM C) +1 and 4.8 BM D) +2 and 4.89 BM
38. The increasing value of the magnetic moment of the following species is
 I. $[\text{Fe}(\text{CN})_6]^{4-}$ II. $[\text{Fe}(\text{CN})_6]^{3-}$ III. $[\text{Cr}(\text{NH}_3)_6]^{3+}$ IV. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 A) I < II < III < IV B) IV < III < II < I
 C) II < III < I < IV D) I < II < IV < III
39. In the coordination compound, $\text{Na}_2[\text{Pt}(\text{CN})_4]$ the Lewis acid is
 A) $[\text{Pt}(\text{CN})_4]^{2-}$ B) Na^+ C) Pt^{2+} D) CN^-
40. The IUPAC name of $[\text{Cu}(\text{NH}_3)_4(\text{NO}_3)_2]$ is:
 A) Tetraamminedinitrito -N-copper (II)
 B) Tetraamminedinitrito -O-copper (II)
 C) Tetraamminedinitrato copper (II)
 D) Tetraamminedinitrido copper (II)
41. Which of the following shows maximum molar conductance?
 A) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ B) $[\text{Co}(\text{NH}_3)_3]\text{Cl}_3$
 C) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ D) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
42. Match Column-I with Column-II and select the correct answer with respect to hybridisation using the codes given below:

	Column - I (Complex)		Column - II (Hybridisation)
(I)	$[\text{AuF}_4]^-$	(p)	dsp^2 hybridisation
(II)	$[\text{Cu}(\text{CN})_4]^{3-}$	(q)	sp^3 hybridisation
(III)	$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$	(r)	sp^3d^2 hybridisation
(IV)	$[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$	(s)	d^2sp^3 hybridisation

Codes:

(I) (II) (III) (IV)

A) q p r s

C) p q r s

(I) (II) (III) (IV)

B) p q s r

D) q p s r

43. Which of the following statements is not correct?

A) $La(OH)_3$ is less basic than $Lu(OH)_3$.

B) In lanthanide series ionic radius of Ln^{3+} ions decreases.

C) La is actually an element of transition series rather than lanthanide series.

D) Atomic radii of Zr and Hf are same because of lanthanide contraction.

44. Which oxide of Mn is acidic in nature?

A) MnO

B) Mn_2O_7

C) Mn_2O_3

D) MnO_2

45. Among the following the compound that is both paramagnetic and coloured is:

A) $K_2Cr_2O_7$

B) $(NH_4)_2[TiCl_6]$

C) $VOSO_4$

D) $K_3[Cu(CN)_4]$

46. Match the following

Column – I	Column – II
A) Lindlar catalyst	1) Hydrogenation of oils
B) Fenton's reagent	2) $Pd + BaSO_4$
C) Ziegler natta catalyst	3) $FeSO_4 + H_2O_2$
D) Raney nickel	4) $R_3Al + TiCl_4$

A) A-2, B-3, C-4, D-1

B) A-2, B-4, C-3, D-1

C) A-1, B-2, C-3, D-4

D) A-4, B-1, C-2, D-3

47. The actinoids exhibit more number of oxidation states in general than the lanthanoids.

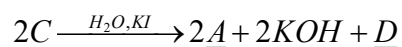
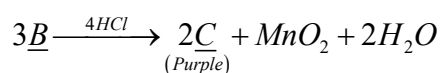
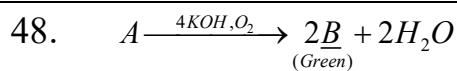
This is because

A) The actinoids are more reactive than the lanthanoids.

B) The 5f orbitals extend farther from the nucleus than the 4f orbitals.

C) The 5f orbitals are more buried than the 4f orbitals

D) There is a similarity between 4f and 5f orbitals in their angular part of the wave function



In the above sequence of reactions, $\underline{A}$ and $\underline{D}$, respectively, are :

A) KIO_3 and MnO_2 B) MnO_2 and KIO_3 C) KI and K_2MnO_2 D) KI and $KMnO_4$

49. The colour of $KMnO_4$ is due to:

A) $M \rightarrow L$ charge transfer transition B) $d-d$ transition
C) $L \rightarrow M$ charge transfer transition D) $\sigma-\sigma^*$ transition

50. Identify the incorrect statement among the following.

A) The chemistry of various lanthanoids is very similar.
B) 4f and 5f orbitals are equally shielded.
C) d-block elements show irregular and erratic chemical properties among themselves.
D) La and Lu have partially filled d orbitals and no other partially filled orbitals.

SECTION-II (NUMERICAL VALUE ANSWER TYPE)

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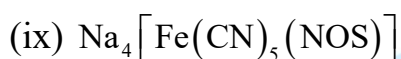
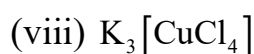
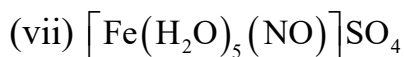
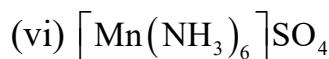
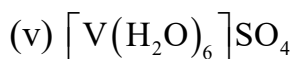
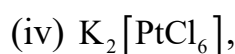
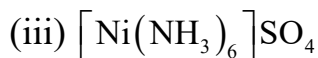
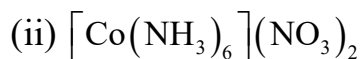
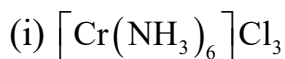
51. Sum of Denticity of following ligands are Glycinate ion, Oxalate ion, o-phenathroline, 2,2-bipyridyl, diethylenetriamine, ethylenediamine

52. How many total sodium ions are present in one formula unit of sodium ethane-1,2 diaminetetraacetatochromate (II) and sodium hexanitrito cobaltate (III)?

53. Blue colour complex is obtained in the analysis of Fe^{+3} having formula $Fe_4[Fe(CN)_6]_3$

Let a = oxidation number of Iron in the coordination sphere b = no. of secondary valencies of central iron ion. c = Effective atomic number of Iron in the coordination sphere. Then find the value of $(c + a - 2b)$

54. Total number of paramagnetic complexes which are inner orbital complexes:



55. The volume (in mL) of 0.1 M AgNO_3 required for complete precipitation of chloride ions present in 30 mL of 0.01 M solution of $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$, as silver chloride is close to

56. EDTA^{4-} is ethylenediaminetetraacetate ion. The total number of $\text{N}-\text{Co}-\text{O}$ bond angles in $[\text{Co}(\text{EDTA})]^{1-}$ complex ion is:

57. Number of equivalent bonds in $\text{Cr}_2\text{O}_7^{2-}$ is X. The value of 10X is _____.

58. The Number of electrons satisfying $n+l=7$ for Lu (Atomic number : 71)

59. How many of the following compounds are diamagnetic and colored ?



60. How many of given statements are true for lanthanums?

(1) The common oxidation state of all lanthanides is +3.

(2) All lanthanides belongs to group III of modern periodic table.

(3) All lanthanides belongs to seventh period of periodic table.

(4) The radius for all lanthanides cations (M^{+3}) decrease from Ce^{+3} to Lu^{+3} .

(5) The basic character of hydroxide increases on moving from $\text{Ce}(\text{OH})_3$ to $\text{Lu}(\text{OH})_3$.

(6) All lanthanides are metals.

(7) The reducing power of lanthanides decreases from Ce to Lu.

(8) Lanthanides reacts with cold water to liberate H_2 .

MATHEMATICS

MAX.MARKS: 100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. Let $f(n) = \sum_{r=0}^n \sum_{k=r}^n \binom{k}{r}$. the total number of divisors of $f(9)$ is equal to.
 A) 4 B) 6 C) 8 D) 10
62. In how many ways can a die be thrown thrice by a person to make a sum of 12
 A) 25 B) 35 C) 21 D) 24
63. Find the number of non-negative integral solutions of $3x + y + z = 24$
 A) 115 B) 114 C) 117 D) 113
64. Number of positive integral solution of $15 < x_1 + x_2 + x_3 \leq 20$ is
 A) 625 B) 635 C) 655 D) 685
65. If N is the number of ways in which a person can walk up a stairway which has 7 steps if he can take 1 or 2 steps up the stairs at a time, then the value of $\frac{N}{3}$ is
 A) 7 B) 9 C) 11 D) 13
66. Total number of ways of selecting two numbers from the set $\{1, 2, 3, 4, \dots, 90\}$ so that their sum is divisible by 3, is
 A) 885 B) 1335 C) 1770 D) 3670
67. The number of four letter word that can be formed by the letters of the word **RAMANA** is
 A) 62 B) 72 C) 52 D) 42
68. The number of odd divisors of 128 is
 A) 8 B) 7 C) 0 D) 1
69. The number of non-negative integral solution of $x_1 + x_2 + x_3 + x_4 \leq n$ (where n is a non-negative integer) is
 A) ${}^{(n+3)}C_3$ B) ${}^{(n+5)}C_5$ C) ${}^{(n+4)}C_4$ D) ${}^{(n+6)}C_6$
70. The number of integral solution to the system of equations
 $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ and $x_1 + x_2 = 15$ when $x_k \geq 0$ ($k = 1, 2, 3, 4, 5$) is
 A) 300 B) 350 C) 336 D) 316

71. The number of positive integral solution for the equation $x + y + z + t = 20$, where x, y, z, t are all ≥ -1 is
 A) ${}^{20}C_4$ B) ${}^{23}C_3$ C) ${}^{27}C_4$ d) ${}^{27}C_3$
72. There are 5 different colour balls and 5 boxes of colours same as those of the balls. The number of ways in which one can place the balls into the boxes, one each in a box, so that no ball goes to a box of its own colour is
 A) 40 B) 44 C) 45 D) 60
73. The number of ways of distributing 9 identical balls in 3 distinct boxes so that none of the boxes is empty is
 A) 8C_3 B) 28 C) 3^8 d) 5
74. Four distinct balls b_1, b_2, b_3, b_4 are to be placed in 5 distinct boxes B_1, B_2, B_3, B_4, B_5 not more than one in a box. The number of ways in which they can be arranged so that b_i will not go to B_i for any 'i' is $D(n)$ denotes de-arrangement
 A) $D(5)$ B) $5D(4)$ C) $D(5) + D(4)$ D) $D(5) + 5D(4)$
75. Equation of the parabola whose axis $y = x$, denotes of vertex from the origin is $\sqrt{2}$ and the distance of focus from the origin is $2\sqrt{2}$ (vertex and focus lie in the 1st quadrant) is
 A) $(x + y)^2 = 2(x + y - 2)$ B) $(x - y)^2 = 8(x + y - 2)$
 C) $(x - y)^2 = 4(x + y - 2)$ D) $(x + y)^2 = 4(x + y - 2)$
76. If 'K' balls are arranged in circular order, then what is the number of ways of selecting four of the balls such that no two of which are consecutive?
 A) $\frac{k(k-7)(k-6)(k-5)}{4!}$ B) $\frac{k(k-7)(k-6)(k-5)(k-4)}{4!}$
 C) $\frac{k(k-7)(k-6)}{4!}$ D) $\frac{(k-5)(k-6)(k-7)}{3!}$
77. In how many ways a bill of 10 Rs paid by three students Ram, Shyam and Ramana if they have 6 Rs, 7 Rs, 8 Rs respectively
 A) 37 B) 57 C) 47 D) 27
78. If $\int \frac{(1 + x \cos x - \sin x) dx}{(x + \sin x + \cos x)^{\frac{1}{2}} (x - \sin x + \cos x)^{\frac{3}{2}}} = (x + l \sin x + m \cos x)^n (x + p \sin x + q \cos x)^r + C$
 (where C is constant of integration and $l, n > 0$) then the value of $\left(\frac{1+m}{n}\right)(p-q)r$ is equal to ____
 A) 2 B) 4 C) 6 D) 8

79. Let A,B,C,D are four coplanar points in 3D-space and $3\overline{OA} + 2\overline{OB} - 4\overline{OC} + \lambda\overline{OD} = \vec{0}$ then the ratio made by $\overline{AB}$ on the joint of $\overline{CD}$ is
A) 5:4 B) 1:4 C) 1:1 D) 4:5
80. No. of different garlands that can be made using 5 flowers of one kind, 3 flowers of another kind is ____
A) 4 B) 8 C) 6 D) 5

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let the product of all the divisors of 1440 be P. If P is divisible by $(24)^k$, then the number of positive integral divisors of maximum value of k is _____
82. Let $x_1, x_2, x_3, \dots, x_k$ are the total divisors of a positive integer n. If $\sum_{i=1}^k x_i = 93$ and $\sum_{i=1}^k \frac{1}{x_i} = \frac{93}{50}$, then the value of 'k' is _____
83. Number of positive integral solution of the equation $x_1 x_2 = 432$
84. Find the number of ways in which 3 distinct numbers can be selected from the set $\{3^1, 3^2, 3^3, \dots, 3^{100}, 3^{101}\}$ so that thefrom G.P.
85. If the number of integral solutions of the equation $2x + 2y + z = 20$ where $x \geq 0, y \geq 0$ and $z \geq 0$ is 11k then k=____
86. Let N represents the number of ways in which 3 numbers a, b & c can be selected from first 100 natural numbers, such that $\vec{p} = a\vec{i} + b\vec{j} + c\vec{k}$ is parallel to $\vec{q} = \vec{i} + 2\vec{j} + 4\vec{k}$, then the value of N
87. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = -x^3 - 3x^2 - 6x + 1$. Number of integers in the solution set if x satisfying the inequality $f(f(x^3 + f(x))) \geq f(f(-f(x) - x^3))$ is ____
88. $\vec{U} = i + j + k, \vec{V} = ai + bj + ck$ ($\vec{V}$ is non zero vector) (Where i, j, k are unit orthogonal vectors) Here $a, b, c \in \{-2, -1, 0, 1, 2\}$ Total number of Possible $\vec{V}$, such that $\vec{U} \cdot \vec{V} = 0$.
89. The number of non-zero solution of the equation $x^2 - 5x - (\text{sgn } x)6 = 0$ is ____
90. The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\sum_{t=0}^{r-1} \frac{1}{7^n} \cdot {}^n C_r \cdot {}^r C_t \cdot 5^t \right)$ is equal to _____